AUTINSIGHT · TECHNICAL REFERENCE

Explainable AI-Driven Pediatric ASD Screening System

Project architecture, datasets, model development, evaluation, explainability, API, user workflow and reproducibility notes. Prepared as a technical companion to the caregiver-facing prototype.

CARE GIVER UI	NEXT.JS / REACT
Questionnaire + observations	Browser state + sessionStorage
API request	POST /predict
FASTAPI SERVICE	PYDANTIC + MODEL RUNTIME
Input validation	Saved preprocessing + XGBoost
Structured response	Class, message, SHAP contributions

Academic prototype only. This application is not a diagnostic system, clinical decision-support tool, validated questionnaire administration, or substitute for a qualified professional. Its primary UCI target is derived from the same ten questionnaire scores given to the model.

Project folder: **files-pasted-by-the-user-master** · Technical documentation is also maintained in the project `docs/` directory.

1. Project purpose and scope

AutInsight combines a caregiver-oriented local web application with a reproducible tabular machine-learning study. The research components audit source data, compare conventional classifiers, preserve the primary model artifact, and return a local feature attribution. The public-facing flow presents a questionnaire, a structured caregiver context assessment, a summary, practical next steps, and a separate visual explainer page.

The design deliberately distinguishes a model class from a clinical conclusion. The screen adapts ten AQ-10 Child-Version-informed items for readability. The wording and digital experience are not validated. Caregiver observations are collected to help prepare for a clinician conversation; only age and ten scored items are sent to the inference API.

Key implementation boundary

The backend is authoritative for schema validation, preprocessing, inference, and per-case SHAP output. The frontend presents returned data but does not reproduce model inference. The web application does not store submissions on the server. Caregiver context remains in browser tab sessionStorage for the summary.

System components

Layer	Technology / location	Responsibility
Frontend	Next.js 16 App Router, React 19, TypeScript (`frontend/`)	Home, screening, summary, visual explanation, print views; request to configured API.
Backend	FastAPI, Pydantic (`backend/app/`)	Health/readiness, input validation, artifact loading, prediction, response and CORS.
Primary artifact	XGBoost pipeline (`models/primary.joblib`)	Fitted preprocessing and classifier, version manifest, and TreeSHAP explainer.
Research scripts	Pandas, scikit-learn, XGBoost, SHAP (`scripts/`)	Data acquisition/audits, EDA, model fitting and metrics generation.
Outputs	`experiments/figures`, `experiments/results`, `experiments/reports`	Dataset-level figures, evaluation metrics, audit reports.

2. Application architecture and flow

The browser uses the Next.js interface at port 3000. Its `NEXT_PUBLIC_API_URL` points directly to FastAPI at port 8000. CORS uses `ALLOWED_ORIGINS` for exact allowed browser origins. Local defaults support both `http://localhost:3000` and `http://127.0.0.1:3000`.

Step	What happens
1 · Start	The home page explains the prototype and leads into screening.
2 · Screening	Caregiver answers ten adapted multiple-choice questions and provides age.
3 · Assessment	Caregiver answers duration, settings, discussion areas, strengths, regression, and prior-support prompts.
4 · Inference	Frontend posts age, categorical context defaults, and ten binary question scores to `/predict`. Context-only observations are not transmitted.
5 · Results	UI shows the dataset label interpretation, response score, local feature contributions, observations, disclaimer, and recommended next steps.
6 · Understand	A separate route visualizes the current TreeSHAP contributions and shows generated research figures and explanations of evaluation methods.

Request and response

`POST /predict` accepts `age`, `gender`, `jaundice`, `family_asd`, and `questions` (10 binary values). Optional metadata fields have defaults. Pydantic rejects invalid input with HTTP 422. The response includes a screening label, probability, reliability tier, message, explanation contributions, explanation scale, limitations and model version. The caregiver interface does not present the raw probability as a clinical risk estimate.

Operational endpoints

Route	Use
GET /health	Process liveness.
GET /ready	Model artifact/readiness check.
GET /model	Model name, version and readiness metadata.
GET /research/metrics	Generated UCI model metrics JSON.
GET /research/indian-metrics	Generated Indian cohort metrics JSON.
POST /predict	Validated inference and optional local SHAP explanation.

No authentication, rate limiting, persistent request database, or formal retention policy is configured. Avoid real patient information and identifiers. Hosting logs may contain operational metadata.

3. Datasets, targets and audit decisions

The two datasets answer different research questions and are never pooled or treated as directly comparable.

Primary UCI child screening dataset

The project uses the UCI Machine Learning Repository Autism Screening on Children dataset (dataset 419; DOI 10.24432/C5659W; CC BY 4.0). Source contains 292 records; two exact duplicate rows are removed for analysis, leaving n=290. The saved holdout experiment uses 217 training and 73 test records. `result`, constant `age_desc`, and identifiers are excluded from model features; train-fitted imputers, encoders and numeric scaling are stored with the pipeline.

Target circularity: the audit verified for every row that `result = sum(A1…A10)` and `class = YES` exactly when `result ≥ 7`. Consequently, models using questionnaire inputs learn a label derived from those same inputs. High scores estimate label reconstruction, not clinical autism detection.

Secondary Indian AIIMS modified INDT-ASD validation cohort

The separately analyzed PLOS ONE 2019 S2 workbook contains 225 deidentified records from a North Indian tertiary referral hospital study. The target maps source code 1 to ASD and 2 to non-ASD; one 180-month age is outside the paper's stated 1–14-year scope and excluded, yielding 224 records and a 168/56 train/test split. Features include 28 instrument items, age in months, gender, SES and residence. IDs, diagnostic summaries, instrument outcomes and other outcome proxies are excluded to reduce leakage.

The Indian cohort uses expert-consensus DSM-5 diagnosis as its target, unlike the UCI questionnaire-rule label. It is small and selected from suspected referrals; it does not represent community screening.

Data handling

Acquisition scripts record source and checksum/provenance. Raw data is git-ignored. Validation scripts report schema, missingness, duplicates, target logic, and sample counts. EDA figures describe source cohorts. Missing values are imputed inside train-fitted sklearn pipelines, so holdout data does not determine imputation values.

Sources

UCI dataset: https://archive.ics.uci.edu/dataset/419/autistic+spectrum+disorder+screening+data+for+children · PLOS ONE 2019: https://doi.org/10.1371/journal.pone.0213242 · AQ-10 Child source guide: https://northdevoncoastalpcn.nhs.uk/media/hypg30xa/aq-10child-autism.pdf

4. Model development and evaluation

Both dataset experiments use a single stratified 75/25 train/test split with random seed 17. Preprocessing is fit on training records only. This simple holdout design is exploratory and produces substantial uncertainty, especially for the 56-record Indian test sample.

Dataset / model	Accuracy	Balanced accuracy	F1	ROC AUC	Brier
UCI · Logistic Regression	0.973	0.974	0.972	1.000	0.019
UCI · Random Forest	0.945	0.944	0.941	0.989	0.094
UCI · XGBoost (primary)	0.973	0.973	0.971	0.998	0.037
Indian · Logistic Regression	0.857	0.849	0.879	0.945	0.088
Indian · Random Forest	0.893	0.880	0.912	0.967	0.073

UCI full-feature model metrics are inflated by label construction. The explicit rule `sum(A1…A10) ≥ 7` agrees with UCI class labels by definition. The demographics-only logistic model is included as a sensitivity analysis (accuracy 0.534; ROC AUC 0.583), but this does not make the full-feature results diagnostic evidence.

Metric interpretation

Measure	Meaning
Accuracy	Fraction of holdout predictions matching recorded target.
Balanced accuracy	Average recall over target classes, reducing the effect of class imbalance.
F1	Harmonic mean of precision and recall for the positive class.
ROC AUC	Ranking discrimination across decision thresholds; not calibration.
Brier score	Mean squared error of predicted probabilities against binary targets; depends on target validity and calibration.

Limitations of the experiment

No repeated/nested cross-validation, prospective evaluation, independent community cohort, confidence interval for model-wide performance, calibration procedure, clinical decision threshold, or external validation is available. Performance on the UCI target cannot establish clinical performance. Indian results are from a selected tertiary-care suspected-case cohort and should not be interpreted as population screening performance.

5. Explainable AI and what the plots mean

Local TreeSHAP

SHAP (SHapley Additive exPlanations) assigns feature contributions to a model output using Shapley-value ideas. TreeSHAP computes these attributions efficiently for tree ensembles. The current results page and “Understand this summary in detail” view use the primary XGBoost local contributions returned by the API. Horizontal bars show signed movement toward or away from the model’s UCI YES class, scaled within that one prediction. They do not show clinical impact, causal effects, or risk percentages.

The primary artifact contains a TreeSHAP explainer and API local contributions. The current project has not independently verified XGBoost additivity, explanation stability, perturbation fidelity, clinical plausibility or caregiver comprehension. API may omit explanations if the explainer is unavailable.

Global SHAP

For the comparative UCI Random Forest, research outputs include mean absolute SHAP feature importance and a local held-out example. Mean absolute values aggregate magnitude across records and indicate model reliance. Questionnaire items lead, consistent with the UCI target rule. RF SHAP additivity for the example was checked to absolute error 1.89×10■¹■. That check is specific to the RF model/example and does not validate the XGBoost explanation or clinical meaning.

Research figures

The app explainer route displays generated descriptive plots: questionnaire response means, source class counts, cohort age distributions, and missingness for UCI and Indian data, plus Indian target distribution. These describe research datasets, not the individual caregiver result. Model metrics, subgroup summaries, provisional reliability bins, and dataset analysis remain in generated JSON/docs for research review.

Methods intentionally not claimed

No counterfactual generation is implemented. No calibrated reliability diagram, entropy score, Jensen–Shannon divergence, clinician-approved threshold, or fairness guarantee is provided. Subgroup metrics with Wilson intervals are descriptive; small group sizes make them unsuitable for strong conclusions. The Indian and UCI studies are not pooled.

Interpretation rule

An explanation describes how a fitted model produced an output under its learned patterns. It does not explain why a child behaves a certain way. Because UCI’s label is a score threshold on the same items, SHAP values largely trace the model’s reconstruction of this score-derived class.

6. Software structure and reproducibility

Path	Contents
`frontend/src/app/`	Pages: home, screening, results, results/understand; shared layout/styles.
`frontend/src/lib/questionnaire.ts`	Question text, response options, feature display mappings and score transformation.
`backend/app/api/`	API route definitions and HTTP behavior.
`backend/app/schemas/`	Pydantic request/response models.
`backend/app/services/runtime.py`	Artifact load, preprocessing/model inference and local explanation runtime.
`backend/app/core/config.py`	Environment settings, including model path and CORS origins.
`scripts/`	Data downloads, audits, EDA, model training.
`docs/`	Architecture, methods, model card, data dictionary, limitations, setup manual.
`models/`	Primary trained model and artifact manifest.
`experiments/`	Generated figures, reports and metrics; raw source files are excluded.

Local runtime requirements

Python 3.11/3.12 and Node.js ≥20.19 (Node 22 recommended). On macOS, XGBoost requires a compatible OpenMP runtime such as Homebrew `libomp`. The teammate guide `docs/TEAMMATE_SETUP.md` gives copy/paste setup for two terminals. Running the app uses the included `models/primary.joblib`; research data downloads and training scripts are optional and not needed for normal inference.

Reproduction outline

Install Python requirements; acquire UCI and Indian source files using their individual download scripts; run dataset validators and EDA scripts; train UCI and Indian experiments separately; inspect generated metrics; and update the model artifact only as an intentional experiment. Fix the seed and preserve provenance/checksums when comparing runs. Do not overwrite source data or mix target semantics.

Security and privacy posture

No authentication is configured. The browser sends age and questionnaire scores to the configured API. Caregiver context resides in current-tab sessionStorage only. Never use this prototype with identifiable information or real patient records. Treat the serialized joblib model as trusted local code; load only project artifacts.

References

Lundberg & Lee, “A Unified Approach to Interpreting Model Predictions,” NeurIPS 2017, https://papers.nips.cc/paper/7062-a-unified-approach-to-interpreting-model-predictions · SHAP documentation, https://shap.readthedocs.io/ · Project reports: `docs/methodology.md`, `docs/model_card.md`, `docs/data_dictionary.md`, `docs/reproducibility.md`, `experiments/results/model_metrics.json`, `experiments/results/indian_model_metrics.json`.

Clinical note: If a caregiver has developmental concerns, they should speak with a qualified pediatric or child-development professional regardless of this prototype’s output. The project is not a diagnosis and cannot rule autism in or out.